

PROFESSIONAL SUMMARY

AI infrastructure engineer with nearly five years of cloud-native AI/HPC experience, focused on production LLM serving since 2024. Owns a distributed inference runtime for heterogeneous accelerator platforms, a Rust KV-aware Router, a cluster-local load-aware Planner, and a production batch-inference platform; contributes to a Kubernetes-native MaaS control plane, the KV Event pipeline, and research and integration for a unified gateway. Strong in Go and Python, with Rust engineering experience across model lifecycle, request routing, capacity management, and recovery.

EXPERIENCE

Zhejiang Lab Aug 2021 - Present

Senior Research Specialist · High-Performance Computing Facility Research Center 2024 - Present

LLM Serving / Model-as-a-Service (MaaS) Platform

- Contributed to the architecture and model-service CRDs of a Kubernetes-native MaaS control plane, expressing models, resource profiles, and service forms through one declaration for template-based one-click deployment; implemented inference-instance pools, status and endpoint aggregation, and supported production releases and diagnosis.
- Owned an in-house distributed inference runtime, implementing the SGLang engine adapter, inference-instance registration and discovery, and lifecycle management while unifying model publication, topology coordination, layered readiness, and traffic draining.
- Owned a cluster-local load-aware Planner and Rust KV-aware Router: combined live load, queues, KV state, and offline capacity profiles for replica planning, then routed requests from prefix KV overlap, inference-instance load, and health.
- Independently designed and implemented a production offline batch-inference platform; contributed research and integration work for a unified gateway, including model onboarding, configuration validation, and cross-component production diagnosis.

Engineering Specialist · Intelligent Supercomputing Research Center Aug 2021 - 2023

Cloud-Native AI / HPC Platform

- Customized Kubernetes Device Plugin and resource-management paths for heterogeneous accelerator discovery, node resource reporting, container-runtime validation, device-level allocation, and fault isolation.
- Adapted PyTorch workloads, models, and distributed job environments across enterprise Linux distributions and heterogeneous accelerator stacks, establishing the platform foundation for later serving and scheduling work.

SELECTED PRODUCTION SYSTEMS

Model-Serving Control Plane and Distributed Inference Runtime Declarative Serving · Runtime · Readiness

- Contributed to unified model-service CRDs and controllers that express models, resource profiles, service forms, and access endpoints through declarative templates for one-click deployment; the control plane owns desired state while the runtime publishes actual service capability.
- Owned the SGLang engine adapter, inference-instance registration and discovery, and model publication, unifying topology coordination, layered readiness, and lifecycle management while isolating engine internals from the platform.
- Admits newly scaled instances only after complete readiness and registration; on scale-in, drains new traffic before in-flight completion and endpoint unregistration, while isolating failed instances and reducing manual configuration and cross-component diagnosis.

KV-aware Router and KV Event Pipeline Rust Routing · KV Event · Prefix Index

- Implemented a Rust KV-aware Router that selects inference instances within a single-model, single-accelerator routing domain from prefix KV overlap, real-time load, queue state, and health, with SGLang HTTP and gRPC integration.
- Owned the KV Event adapter, ZMQ-to-NATS bridge, Router event consumption, and prefix index, normalizing BlockStored, BlockRemoved, and AllBlocksCleared into Stored, Removed, and Cleared RouterEvents.
- Balances cache locality against live load from the continuously updated prefix index, removes unhealthy inference instances from the candidate set, and supports stable request reselection and scheduling.

SELECTED PRODUCTION SYSTEMS (CONTINUED)

Cluster-Local Load-Aware P/D Planner

Capacity Modeling · Replica Autoscaling · Observability

- Aggregates Router, inference-instance, and Prometheus signals within a cluster, combining request load, queue depth, KV state, and offline capacity profiles to continuously compute service replica targets.
- Plans prefill and decode replicas independently for P/D-disaggregated services and a unified replica target for co-located services; the Planner owns capacity decisions while the control plane and Kubernetes create and place resources.
- In production, scales successfully under pressure to reduce queue backlog and restore stable requests, then reclaims idle replicas as load falls to improve cluster utilization.

Multi-Tenant Offline Batch-Inference Platform

OpenAI Batch API · Scheduling · Recovery

- Built the OpenAI Batch API, streaming JSONL processing, task sharding, concurrency controls, and output/error aggregation for large, long-running data-distillation, generation, and evaluation workloads.
- Persists task progress and uses Kubernetes Lease leadership plus shard reassignment so schedulers or executors can fail without restarting an entire batch.
- Uses at-least-once backend execution with result-level de-duplication and custom_id-based skipping, providing request-level error isolation and stable long-running operation.

CORE TECHNOLOGIES

Runtime & Data Plane: distributed inference runtime, SGLang, KV Event Plane, ZMQ/NATS, P/D disaggregation

Control Plane & Scheduling: Go, Kubernetes Operator, CRD/controller, inference-instance pools, capacity modeling, P/D replica autoscaling, Device Plugin, Helm, ArgoCD

Routing & Gateway: Rust, KV-aware routing, HTTP/gRPC routing, prefix index, LiteLLM research and integration, OpenAI/Anthropic APIs, authentication and rate-limit configuration

Batch & Observability: Python, OpenAI Batch API, Redis, MySQL, Kubernetes Lease, Prometheus, QPS/TTFT/TPOT, request tracing

EDUCATION

Arizona State University - M.S. in Computer Science, GPA 3.97 / 4.00, May 2019 - Dec 2020

Arizona State University - B.S. in Computer Science, Aug 2015 - May 2019

PUBLICATION & INTELLECTUAL PROPERTY

Prompt2Length: A Lightweight Model for Predicting Output Length in LLM Inference - third author, IEEE AIAHPC 2025; co-inventor on two granted invention patents and named contributor on two registered software copyrights.

LANGUAGES

Mandarin Chinese (native); English (full professional proficiency).